In [1]:
```
import pandas as pd
```

In [3]:
```
# TASK 1: Spotify - Remove Duplicate Songs

spotify = pd.read_csv("spotify_streams.csv")
```

In [4]:
```
spotify_cleaned = spotify.drop_duplicates(subset=["song"],keep="first")

print("TASK 1 - Spotify Data After Removing Duplicate Songs")
print(spotify_cleaned)
```

```
TASK 1 - Spotify Data After Removing Duplicate Songs
              song            artist     streams        date
0  Blinding Lights        The Weeknd  4200000000  2026-01-01
1     Shape of You        Ed Sheeran  3800000000  2026-01-02
2       Levitating          Dua Lipa  2100000000  2026-01-03
3          Perfect        Ed Sheeran  1900000000  2026-01-04
5         Believer  Imagine Dragons  1800000000  2026-01-06
6        As It Was     Harry Styles  1700000000  2026-01-07
8             Stay     The Kid LAROI  1600000000  2026-01-09
9          Peaches     Justin Bieber  1500000000  2026-01-10
```

In [5]:
```
# TASK 2: Flipkart - Duplicate Reviews
reviews = pd.DataFrame({"product_id": ["P101", "P102", "P101", "P103","P102", "P
"user_id": ["U01", "U02", "U01", "U03","U02", "U04", "U03"],
"rating": [5, 4, 5, 3, 4, 5, 3],
"review_text": ["Excellent product","Good quality","Excellent product","Average
duplicates_before = reviews.duplicated(subset=["product_id", "user_id"]).sum()

print("\nTASK 2 - Duplicate Reviews Before drop_duplicates()")
print(duplicates_before)
```

```
TASK 2 - Duplicate Reviews Before drop_duplicates()
3
```

In [6]:
```
reviews_cleaned = reviews.drop_duplicates(subset=["product_id", "user_id"],keep=
duplicates_after = reviews_cleaned.duplicated(subset=["product_id", "user_id"]).

print("TASK 2 - Duplicate Reviews After drop_duplicates()")
print(duplicates_after)
```

```
TASK 2 - Duplicate Reviews After drop_duplicates()
0
```

In [8]:
```
# TASK 3: Zomato - IQR Outliers

zomato = pd.DataFrame({"order_id": [
1001, 1002, 1003, 1004, 1005,1006, 1007, 1008, 1009, 1010,1011, 1012],
"order_amount": [250, 320, 450, 280, 510,390, 275, 600, 420, 350,480, 2500]})

Q1 = zomato["order_amount"].quantile(0.25)
Q3 = zomato["order_amount"].quantile(0.75)

IQR = Q3 - Q1
lower_limit = Q1 - 1.5 * IQR
upper_limit = Q3 + 1.5 * IQR

outliers = zomato[(zomato["order_amount"] < lower_limit) |(zomato["order_amount"
```

```
print("\nTASK 3 - Zomato IQR Outliers")
print("Q1:", Q1)
print("Q3:", Q3)
print("IQR:", IQR)
print("Lower Limit:", lower_limit)
print("Upper Limit:", upper_limit)

print("Outlier Order IDs:")
print(outliers["order_id"].tolist())
```

```
TASK 3 - Zomato IQR Outliers
Q1: 310.0
Q3: 487.5
IQR: 177.5
Lower Limit: 43.75
Upper Limit: 753.75
Outlier Order IDs:
[1012]
```

In [9]:

```
# TASK 4: IPL - Z-score Outliers
ipl_sales = pd.DataFrame({"date": ["2026-04-01", "2026-04-02", "2026-04-03","202
"2026-04-07", "2026-04-08", "2026-04-09","2026-04-10", "2026-04-11", "2026-04-12
"tickets_sold": [5200, 5500, 5100, 5300,5400, 5250, 5150, 5350,5450, 5300, 18000


ipl_sales["z_score"] = ((ipl_sales["tickets_sold"] - ipl_sales["tickets_sold"].m

zscore_outliers = ipl_sales[ipl_sales["z_score"].abs() > 2]

print("\nTASK 4 - IPL Ticket Sales Z-score Outliers")
print(zscore_outliers[["date", "tickets_sold", "z_score"]])
```

```
TASK 4 - IPL Ticket Sales Z-score Outliers
          date  tickets_sold   z_score
10  2026-04-11         18000  3.071654
```

In [12]:

```
# TASK 5: Paytm Wallet - Z-score Outlier Function
def find_zscore_outliers(df, column, threshold=2):

    z_scores = (
        (df[column] - df[column].mean())
        / df[column].std()
    )

    return df[z_scores.abs() > threshold]


paytm = pd.DataFrame({
    "transaction_id": [
        "TXN001", "TXN002", "TXN003", "TXN004",
        "TXN005", "TXN006", "TXN007", "TXN008",
        "TXN009", "TXN010"
    ],
    "transaction_amount": [
        250, 300, 275, 320, 290,
        310, 280, 295, 15000, 260
    ]
})

paytm_outliers = find_zscore_outliers(
    paytm,
```

```
    "transaction_amount"
)

print("\nTASK 5 - Paytm Wallet Z-score Outliers")
print(paytm_outliers)
```

```
TASK 5 - Paytm Wallet Z-score Outliers
  transaction_id  transaction_amount
8         TXN009               15000
```

In [ ]:

```
```